DPEN022: Enabling Mathematics B
Solutions to Class Questions Week 5

Some questions only include the final answer and do not have fully worked solutions.

Week 5A

Key Idea – Definite Integrals of Power of x

1. (a)
$$\int_{-1}^{2} x^3\,dx = \left[\frac{x^4}{4}\right]_{-1}^{2} = \frac{2^4}{4} - \frac{(-1)^4}{4} = \frac{15}{4}$$

(b) 0

(c)
$$\int_{-1}^{0} 2x^5 + x - 7\,dx = \left[\frac{2x^6}{6} + \frac{x^2}{2} - 7x\right]_{-1}^{0}$$
$$= \left[\frac{x^6}{3} + \frac{x^2}{2} - 7x\right]_{-1}^{0}$$
$$= \left(\frac{0^6}{3} + \frac{0^2}{2} - 7\times 0\right) - \left(\frac{(-1)^6}{3} + \frac{(-1)^2}{2} - 7\times(-1)\right)$$
$$= 0 - \left(\frac{1}{3} + \frac{1}{2} + 7\right)$$
$$= -\frac{47}{6}$$

(d) $22\frac{11}{12}$

(e)
$$\int_{3}^{4} \frac{5x^4(x+2)}{x^3}\,dx = \int_{3}^{4} \frac{5x^5 + 10x^4}{x^3}\,dx = \int_{3}^{4} \frac{5x^5}{x^3} + \frac{10x^4}{x^3}\,dx$$
$$= \int_{3}^{4} 5x^2 + 10x\,dx = \left[\frac{5x^3}{3} + \frac{10x^2}{2}\right]_{3}^{4}$$
$$= \left[\frac{5x^3}{3} + 5x^2\right]_{3}^{4}$$
$$= \left(\frac{5\times 4^3}{3} + 5\times 4^2\right) - \left(\frac{5\times 3^3}{3} + 5\times 3^2\right)$$
$$= \left(\frac{320}{3} + 80\right) - (45 + 45)$$
$$= \frac{290}{3}$$

(f)

$$\int_{-5}^{5}(x-1)\left(x^2+x+1\right)dx=\int_{-5}^{5}x^3+x^2+x-x^2-x-1\,dx$$
$$=\int_{-5}^{5}x^3-1\,dx$$
$$=\left[\frac{x^4}{4}-x\right]_{-5}^{5}$$
$$=\left(\frac{5^4}{4}-5\right)-\left(\frac{(-5)^4}{4}-(-5)\right)$$
$$=\left(\frac{5^4}{4}-5\right)-\left(\frac{5^4}{4}+5\right)$$
$$=\frac{5^4}{4}-5-\frac{5^4}{4}-5$$
$$=-10$$

(g) $\frac{16}{3}$

(h)

$$\int_{1}^{27}\frac{1}{2\sqrt[3]{x^2}}\,dx=\frac{1}{2}\int_{1}^{27}x^{-\frac{2}{3}}dx$$
$$=\frac{1}{2}\left[\frac{x^{\frac{1}{3}}}{\frac{1}{3}}\right]_{1}^{27}=\frac{1}{2}\left[3\sqrt[3]{x}\right]_{1}^{27}$$
$$=\frac{1}{2}\left(3\sqrt[3]{27}-3\sqrt[3]{1}\right)$$
$$=\frac{1}{2}(3\times3-3\times1)$$
$$=3$$

(i) $-\frac{1}{12}$

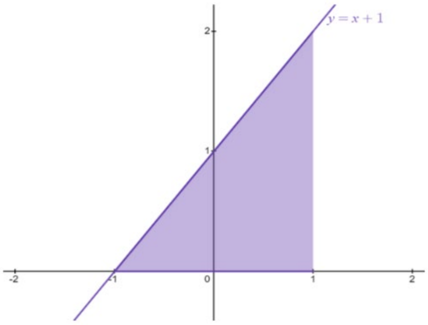
Key Idea – Area under the Curve

1.

$$\text{Area }=\int_{-1}^{2}5x^4+1\,dx$$
$$=\left[\frac{5x^5}{5}+x\right]_{-1}^{2}$$
$$=\left[x^5+x\right]_{-1}^{2}$$
$$=\left(2^5+2\right)-\left((-1)^5+(-1)\right)$$
$$=34-(-2)=36\text{ units}^2$$

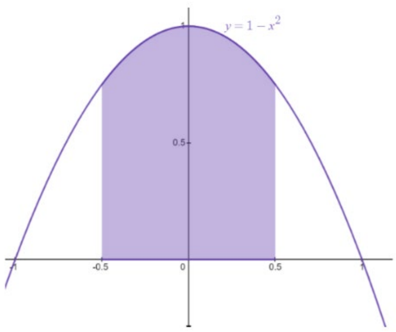
2. (a)

$$\begin{aligned}\text{Area } &= \int_{-1}^{1} x+1\,dx \\ &= \left[\frac{x^2}{2}+x\right]_{-1}^{1} \\ &= \left(\frac{1^2}{2}+1\right)-\left(\frac{(-1)^2}{2}+(-1)\right) \\ &= \frac{3}{2}-\left(-\frac{1}{2}\right)=2 \text{ units}^2\end{aligned}$$



(b)

$$\begin{aligned}\text{Area } &= \int_{-0.5}^{0.5} 1-x^2\,dx \\ &= 2\int_{0}^{0.5} 1-x^2\,dx \quad \text{(since the graph is symmetrical)} \\ &= 2\left[x-\frac{x^3}{3}\right]_{0}^{0.5} \\ &= 2\left(\left(0.5-\frac{0.5^3}{3}\right)-\left((0)-\frac{(0)^3}{3}\right)\right) \\ &= 2\left(\frac{11}{24}-0\right)=\frac{11}{12} \text{ units}^2\end{aligned}$$



(c) $\text{Area } = \int_{1}^{4} x^2-4x+4\,dx = 3 \text{ units}^2$

<u>Week 5B</u>

Key Idea – Area below the *x*-axis and Subdividing Regions before Integrating

1.

$$\begin{aligned}\text{Area } &= -\int_{1}^{3} 1-x^4\,dx \\ &= -\left[x-\frac{x^5}{5}\right]_{1}^{3} \\ &= -\left(\left(3-\frac{3^5}{5}\right)-\left(1-\frac{1^5}{5}\right)\right) \\ &= -\left(-\frac{228}{5}-\frac{4}{5}\right) \\ &= -\left(-\frac{232}{5}\right)=\frac{232}{5} \text{ units}^2\end{aligned}$$

2. $\text{Area } = \left|\int_{0}^{4} 3x^2(x-4)\,dx\right| = \left|\int_{0}^{4} (3x^3-12x^2)\,dx\right| = 64 \text{ units}^2$

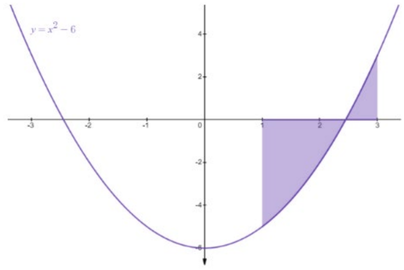
3. (a) $\left(\sqrt{6},\ 0\right)$ and $\left(-\sqrt{6},\ 0\right)$

(b) See below

(c) $8\sqrt{6}$ units^2

4.

First find where the curve crosses the x axis



Let $y=0$ in $y=x^2-6$

$x^2-6=0 \Rightarrow x=\pm\sqrt{6}$

$$\text{Area } = -\int_1^{\sqrt{6}} x^2-6\,dx+\int_{\sqrt{6}}^{3} x^2-6\,dx$$

$$=-\left[\frac{x^3}{3}-6x\right]_1^{\sqrt{6}}+\left[\frac{x^3}{3}-6x\right]_{\sqrt{6}}^{3}$$

$$=-\left(\left(\frac{\left(\sqrt{6}\right)^3}{3}-6\sqrt{6}\right)-\left(\frac{1^3}{3}-6\times1\right)\right)+\left(\left(\frac{3^3}{3}-6\times3\right)-\left(\frac{\left(\sqrt{6}\right)^3}{3}-6\sqrt{6}\right)\right)$$

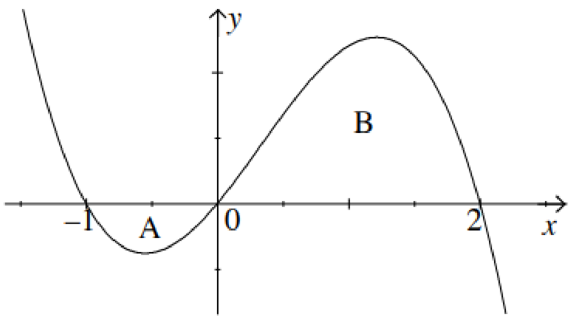
$$=-\left(\frac{6\sqrt{6}}{3}-6\sqrt{6}-\frac{1}{3}+6\right)+\left(9-18-\frac{6\sqrt{6}}{3}+6\sqrt{6}\right)$$

$$=-\frac{6\sqrt{6}}{3}+6\sqrt{6}+\frac{1}{3}-6+9-18-\frac{6\sqrt{6}}{3}+6\sqrt{6}$$

$$=8\sqrt{6}-\frac{44}{3}\text{ units}^2$$

5. The graph of the curve cuts the x-axis at -1, 0 and 2.

The total area = area A + area B.



$$\begin{aligned}\text{Area A} &= \left|\int_{-1}^{0} 12x(x+1)(2-x)dx\right| \\ &= \left|\int_{-1}^{0}(-12x^3+12x^2+24x)dx\right| \\ &= \left|\left[-3^4+4x^3+12x\right]_{-1}^{0}\right| \\ &= |0-(-3-4+12)| \\ &= |-5| = 5.\end{aligned}$$

$$\begin{aligned}\text{Area B} &= \int_0^2 12x(x+1)(2-x)dx \\ &= \left[-3x^4+4x^3+12x^2\right]_0^2 = (-48+32+48) = 32.\end{aligned}$$

Therefore the total area is $5+32=37$ square units.

Key Idea – The Area between two Curve.

1. (i) and (ii) The curves are easier to sketch if we first find the points of intersection: they meet where $x^3 - 7x + 6 = 6 - x - x^2$.

That is,

$$x^3 + x^2 - 6x = 0$$

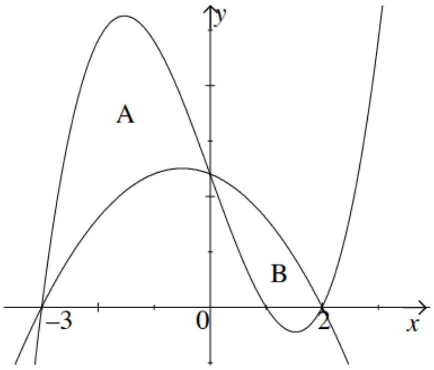
or

$$x(x-2)(x+3) = 0.$$

So the points of intersection are $(0, 6)$; $(2, 0)$; and $(-3, 0)$.

The first curve is an 'upside-down' parabola, and the second a cubic.

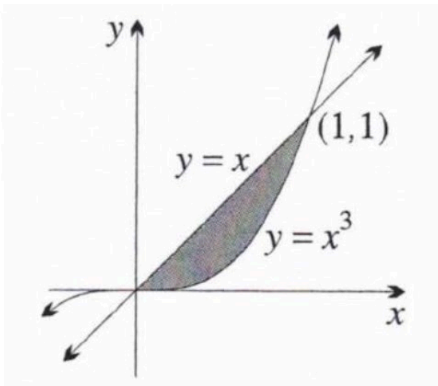
Total area = area A + area B.



$$\begin{aligned} \text{Area A} &= \int_{-3}^{0} ((x^3 - 7x + 6) - (6 - x - x^2))dx \\ &= \int_{-3}^{0} (x^3 + x^2 - 6x)dx \\ &= \left[\frac{1}{4}x^4 + \frac{1}{3}x^3 - 3x^2\right]_{-3}^{0} \\ &= 15\frac{3}{4}. \end{aligned}$$

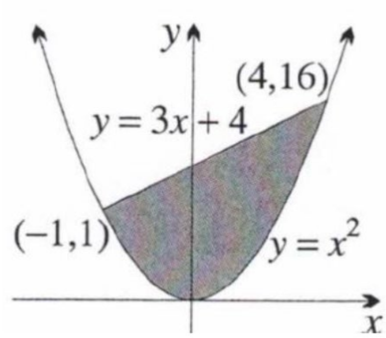
2. (a)

$$\begin{aligned} \text{Area} &= \int_0^1 x - x^3 \; dx \\ &= \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 \\ &= \left(\frac{1^2}{2} - \frac{1^4}{4}\right) - \left(\frac{0^2}{2} - \frac{0^4}{4}\right) \\ &= \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \text{ units}^2 \end{aligned}$$



(b)

$$\begin{aligned} \text{Area} &= \int_{-1}^{4} 3x + 4 - x^2 \; dx \\ &= \left[\frac{3x^2}{2} + 4x - \frac{x^3}{3}\right]_{-1}^{4} \\ &= \left(\frac{3\times(4)^2}{2} + 4\times(4) - \frac{(4)^3}{3}\right) - \left(\frac{3\times(-1)^2}{2} + 4\times(-1) - \frac{(-1)^3}{3}\right) \\ &= \frac{56}{3} - \left(-\frac{13}{6}\right) = \frac{125}{6} \text{ units}^2 \end{aligned}$$



3.

First find where the curves intersect

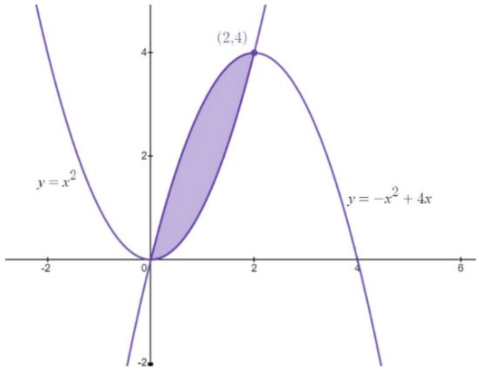
sub $y = x^2$ into $y = -x^2 + 4x$

$x^2 = -x^2 + 4x$

$2x^2 - 4x = 0$

$2x(x-2) = 0$

$x = 0, 2$



$$\text{Area} = \int_0^2 -x^2 + 4x - x^2 \; dx = \int_0^2 -2x^2 + 4x \; dx$$

$$= \left[\frac{-2x^3}{3} + 2x^2\right]_0^2$$

$$= \left(\frac{-2\times 2^3}{3} + 2\times 2^2\right) - \left(\frac{-2\times 0^3}{3} + 2\times 0^2\right)$$

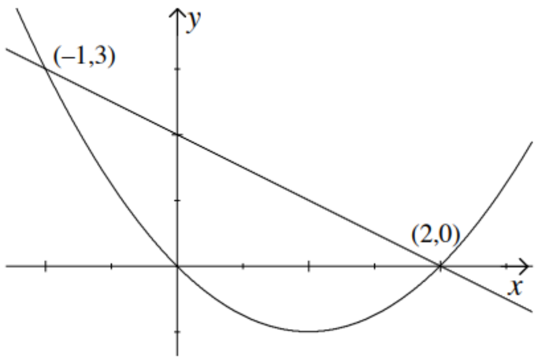
$$= \frac{8}{3} - 0$$

$$= \frac{8}{3} \text{ units}^2$$

4.

The curves $y = x^2 - 2x$ and $y = -x + 2$ intersect where $x^2 - 2x = -x + 2$. i.e. at $x = -1$ or $x = 2$.

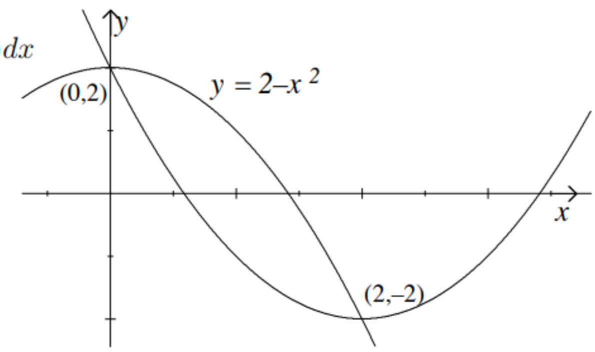
The upper curve is $y = -x + 2$.



$$\begin{aligned} \text{Area} &= \int_{-1}^{2} ((-x+2) - (x^2 - 2x))dx \\ &= \int_{-1}^{2} (2 + x - x^2)dx \\ &= \left[2x + \frac{1}{2}x^2 - \frac{1}{3}x^3\right]_{-1}^{2} \\ &= (4 + 2 - \frac{8}{3}) - (-2 + \frac{1}{2} + \frac{1}{3}) \\ &= 4\frac{1}{2}. \end{aligned}$$

5.

The curves intersect where $x^2 - 4x + 2 = 2 - x^2$ i.e. $2x^2 - 4x = 0$ i.e. $x = 0$ or $x = 2$. The upper curve is $y = 2 - x^2$ (see sketch).



$$\begin{aligned} \text{Area} &= \int_0^2 ((2 - x^2) - (x^2 - 4x + 2))dx \\ &= \int_0^2 (4x - 2x^2)dx \\ &= \left[2x^2 + \frac{2}{3}x^3\right]_0^2 \\ &= (8 - \frac{2}{3}\cdot 8) - 0 \\ &= 2\frac{2}{3}. \end{aligned}$$

Key Idea – Integration of Exponential Functions

1. (a)

$$\int 5e^{5x-3}dx \qquad f(x)=5x$$
$$f'(x)=5$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int 5e^{5x-3}dx = e^{5x-3}+c$$

(b)

$$\int 7e^{7x}+3x^2dx \qquad f(x)=7x$$
$$f'(x)=7$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int 7e^{7x}+3x^2dx = e^{7x}+x^3+c$$

(c) $\int e^x+e^{-3x}dx = e^x-\frac{1}{3}e^{-3x}+c$

(d)

$$\int e^{-x}dx \qquad f(x)=-x$$
$$f'(x)=-1$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int e^{-x}dx = -\int -e^{-x}dx$$
$$=-e^{-x}+c$$

(e)

$$\int e^{3x+4}dx \qquad f(x)=3x+4$$
$$f'(x)=3$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int e^{3x+4}dx = \frac{1}{3}\int 3e^{3x+4}dx$$
$$=\frac{1}{3}e^{3x+4}+c$$

(f)

$$\int \sqrt[3]{e^{5x}}dx = \int \mathrm{e}^{\frac{5x}{3}}dx \qquad f(x)=\frac{5x}{3}$$
$$f'(x)=\frac{5}{3}$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int e^{\frac{5x}{3}}dx = \frac{3}{5}\int \frac{5}{3}e^{\frac{5x}{3}}dx$$
$$=\frac{3}{5}e^{\frac{5x}{3}}+c\left(=\frac{3\sqrt[3]{e^{5x}}}{5}+c\right)$$

(g) $\int e^3dx = e^3x+c$

(h)

$$\int \frac{x}{e^{x^2}}dx = \int xe^{-x^2} \qquad f(x)=-x^2$$
$$f'(x)=-2x$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$
$$\int xe^{-x^2}dx = -\frac{1}{2}\int -2xe^{-x^2}dx$$
$$=-\frac{1}{2}e^{-x^2}+c = -\frac{1}{2e^{x^2}}+c$$

(i) $\int e^{-x}+e\ dx \ = -e^{-x}+ex+c$

(j) $\int \left(e^x+e^{-x}\right)^2\ dx \ = \int \left(e^{2x}+2+e^{-2x}\right)\ dx$
$$=\frac{1}{2}e^{2x}+2x-\frac{1}{2}e^{-2x}+c$$

(k) $\int 5e^{-2.5x}\ dx \ = -2e^{-2.5x}+c$

(l) $\int \frac{e^{2x}+e^{5x}}{e^{2x}}\ dx \ = \int 1+e^{3x}dx$
$$=x+3\frac{1}{\ }e^{3x}+c$$

2. (a)

$$\int_1^2 e^{2x-1}dx \qquad f(x)=2x-1$$
$$f'(x)=2$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+C$
$$\int_1^2 e^{2x-1}dx = \frac{1}{2}\int_1^2 2e^{2x-1}dx$$
$$=\frac{1}{2}\left[e^{2x-1}\right]_1^2$$
$$=\frac{1}{2}\left(e^{2\times2-1}-e^{2\times1-1}\right)$$
$$=\frac{1}{2}\left(e^3-e^1\right)$$

(b)

$$\int_{\ln 3}^{\ln 5} xe^{2x}dx \qquad f(x)=2x$$
$$f'(x)=2$$
Using $\int f'(x)e^{f(x)}dx = e^{f(x)}+C$
$$\int_{\ln 3}^{\ln 5} xe^{2x}dx = \frac{1}{2}\int_{\ln 3}^{\ln 5} 2xe^{2x}dx$$
$$=\frac{1}{2}\left[e^{2x}\right]_{\ln 3}^{\ln 5}$$
$$=\frac{1}{2}\left(e^{2\times\ln 5}-e^{2\times\ln 3}\right)$$
$$=\frac{1}{2}\left(e^{\ln 5^2}-e^{\ln 3^2}\right)$$
$$=\frac{1}{2}\left(5^2-3^2\right)$$
$$=8$$

(c) $\frac{5e^6}{3}-\frac{5e^3}{3}=\frac{5e^3}{3}\left(e^3-1\right)$

(d) $e^{-2}-1=\frac{1}{e^2}-1$

(e)

$$\int_0^{\ln 5} e^x dx = \left[e^x\right]_0^{\ln 5}$$
$$=e^{\ln 5}-e^0$$
$$=5-1$$
$$=4$$

(f)

$$\left(e^{\frac{1}{2}}-e^{-\frac{3}{2}}\right)$$

(g) $\frac{e^2}{2}+\frac{1}{2}e^{-2}-1$

(h) $6e^{\frac{1}{2}}+2e^{-\frac{3}{2}}-2e^{\frac{3}{2}}$

Key Idea – Integration of Logarithmic Functions

1. (a) $\int \frac{1}{x}dx = \ln|x|+C$

(b) $\int \frac{2}{2x-3}dx = \ln|2x-3|+c$

(c) $\int \frac{-9x^2}{2-3x^3}dx = \ln\left|2-3x^3\right|+c$

(d) $\int x+\frac{1}{x-3}dx = \frac{x^2}{2}+\ln|x-3|+c$

(e)

$$\int \frac{3x^2}{x^3+5}dx \qquad f(x)=x^3+5$$
$$f'(x)=3x^2$$
Using $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)|+C$
$$\int \frac{3x^2}{x^3+5}dx = \ln\left|x^3+5\right|+C$$

(f)

$$\int \frac{3}{x}dx = 3\int\frac{1}{x}dx$$
$$=3\ln|x|+C$$

(g) $\int \frac{x}{x^2+2}dx = \frac{1}{2}\ln\left|x^2+2\right|+c$

(h) $\int \frac{x^2}{2x^3-7}dx = \frac{1}{6}\ln\left|2x^3-7\right|+c$

(i)

$$\int \frac{x^3}{x^4+3}\,dx \qquad f(x)=x^4+3$$

$$f'(x)=4x^3$$

Using $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)|+C$

$$\int \frac{x^3}{x^4+3}\,dx = \frac{1}{4}\int \frac{4x^3}{x^4+3}\,dx$$

$$=\frac{1}{4}\ln\left|x^4+3\right|+C$$

(j) $\int \frac{4x}{2x^2+1}\,dx = \ln\left|2x^2+1\right|+c$

(k) $\int \frac{x^2-1}{x}\,dx = \int x-\frac{1}{x}\,dx$

$$=\frac{x^2}{2}-\ln|x|+c$$

(l) $\int\left(x-\frac{1}{x^2}\right)^2 dx = \int\left(x^2-\frac{2}{x}+\frac{1}{x^4}\right)dx$

$$=\int\left(x^2-\frac{2}{x}+x^{-4}\right)dx$$

$$=\frac{x^3}{3}-2\ln|x|+\frac{x^{-3}}{-3}+c$$

$$=\frac{x^3}{3}-2\ln|x|-\frac{1}{3x^3}+c$$

2. (a)

$$\int_0^1 \frac{6x^5}{x^6+1}\,dx \qquad f(x)=x^6+1$$

$$f'(x)=6x^5$$

Using $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)|+C$

$$\int_0^1 \frac{6x^5}{x^6+1}\,dx = \left[\ln\left|x^6+1\right|\right]_0^1$$

$$=\ln\left|1^6+1\right|-\ln\left|0^6+1\right|$$

$$=\ln 2-\ln 1=\ln 2$$

(b) $\frac{1}{2}\ln 5$

(c)

$$\frac{1}{2}(\ln 35-\ln 15)=\frac{1}{2}\ln\left(\frac{35}{15}\right)$$

$$=\frac{1}{2}\ln\left(\frac{7}{3}\right)$$

(d)

$$\int_{e^2}^{e^7}\frac{5}{x}\,dx = 5\int_{e^2}^{e^7}\frac{1}{x}\,dx$$

$$=5\left[\ln|x|\right]_{e^2}^{e^7}$$

$$=5\left(\ln e^7-\ln e^2\right)$$

$$=5(7-2)$$

$$=25$$

(e)

$$\int_1^2 \frac{12}{12x+1}\,dx \qquad f(x)=12x+1$$

$$f'(x)=12$$

Using $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)|+c$

$$\int_1^2 \frac{12}{12x+1}\,dx = \left[\ln|12x+1|\right]_1^2$$

$$=\left(\ln|12\times 2+1|-\ln|12\times 1+1|\right)=\left(\ln 25-\ln 13\right)$$

(f) $\ln 2+\frac{1}{4}$

Key Idea – Integration of Trigonometric Functions

1. (a) $\int 2\sin(2x-1)\,dx = -\cos(2x-1)+c$

(b) $\int 3\sec^2(3x)-x\ dx = 3\tan(3x)-\frac{x^2}{2}+c$

(c) $\int \frac{1}{3}\cos\left(\frac{x}{3}\right)+5\ dx = \sin\left(\frac{x}{3}\right)+5x+c$

(d) $\int \sin(2x+5)\,dx = -\frac{1}{2}\cos(2x+5)+c$

(e) $\int \cos(3x)\,dx = \frac{1}{3}\sin(3x)+c$

(f) $\int \sec^2(7x-1)\,dx = \frac{1}{7}\tan(7x-1)+c$

(g) $\int 6\sin\left(x+\frac{\pi}{4}\right)dx = -6\cos\left(x+\frac{\pi}{4}\right)+c$

(h)
$$\int \frac{\cos(2x+1)}{2}dx = \frac{1}{2}\int \cos(2x+1)\,dx$$
$$=\frac{1}{2}\times\frac{1}{2}\sin(2x+1)+c$$
$$=\frac{1}{4}\sin(2x+1)+c$$

(i)
$$\int 3\sec^2\left(3-\frac{\pi}{2}x\right)dx = 3\int \sec^2\left(3-\frac{\pi}{2}x\right)dx$$
$$=3\times\frac{1}{-\frac{\pi}{2}}\tan\left(3-\frac{\pi}{2}x\right)+c$$
$$=-\frac{6}{\pi}\tan\left(3-\frac{\pi}{2}x\right)+c$$

2. (a) 2 (b) $\sqrt{3}$ (c) 2

(d)
$$\int_0^{\frac{\pi}{3}} 2\sin(3x)\,dx = \left[2\times\left(-\frac{1}{3}\right)\cos(3x)\right]_0^{\frac{\pi}{3}}$$
$$=\left[-\frac{2}{3}\cos(3x)\right]_0^{\frac{\pi}{3}}$$
$$=-\frac{2}{3}\cos\left(3\times\frac{\pi}{3}\right)-\left(-\frac{2}{3}\cos(3\times 0)\right)$$
$$=-\frac{2}{3}\cos(\pi)+\frac{2}{3}\cos(0)$$
$$=-\frac{2}{3}\times(-1)+\frac{2}{3}\times 1=\frac{4}{3}$$

(e)
$$\int_{-\pi}^{0} 3\cos(x+\pi)\,dx = \left[3\sin(x+\pi)\right]_{-\pi}^{0}$$
$$=3\sin(0+\pi)-3\sin(-\pi+\pi)$$
$$=3\sin(\pi)-3\sin(0)$$
$$=3\times 0-3\times 0$$
$$=0$$

(f)

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 4\sec^2 x \, dx = \left[4\tan x\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$$

$$= 4\tan\frac{\pi}{4} - 4\tan\frac{\pi}{6}$$

$$= 4\times 1 - 4\times\frac{1}{\sqrt{3}}$$

$$= 4 - \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}-4}{\sqrt{3}}$$

(g) -0.25 (h) $\frac{\sqrt{2}-1}{2}$ (i) $\frac{-3}{8}$

Key Idea - Integration Mixed

1. (a)

$$\int e^x - \sin x + x + 2 \, dx = e^x + \cos x + \frac{x^2}{2} + 2x + c$$

(b)

$$\int \cos x - \sec^2 x + \frac{1}{x} dx = \sin x - \tan x + \ln x + C$$

(c)

$$\int \sqrt{x} - e^{2x-1} + \sin(2x) dx = \int x^{\frac{1}{2}} - e^{2x-1} + \sin(2x) dx$$

$$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{1}{2}e^{2x-1} - \frac{1}{2}\cos(2x) + C$$

$$= \frac{2\sqrt{x^3}}{3} - \frac{1}{2}e^{2x-1} - \frac{1}{2}\cos(2x) + C$$

(d)

$$\int \sin(-\pi x + 2) + \frac{x^2}{x^3+1} dx = \int \sin(-\pi x + 2) dx + \int \frac{x^2}{x^3+1} dx$$

$$= \frac{1}{-\pi} \times \left(-\cos(-\pi x + 2)\right) + \frac{1}{3}\int \frac{3x^2}{x^3+1} dx$$

$$= \frac{\cos(-\pi x + 2)}{\pi} + \frac{1}{3}\ln\left|x^3+1\right| + C$$

(e)

$$\int 4\cos(2x-1) + 7e^{5x-2} dx = 4\int \cos(2x-1) dx + 7\int e^{5x-2} dx$$

$$= 4\times\frac{1}{2}\sin(2x-1) + \frac{7}{5}\int 5e^{5x-2} dx$$

$$= 2\sin(2x-1) + \frac{7}{5}e^{5x-2} + C$$

(f) $\int\left(\cos\frac{x}{2} - 2\sin 2x\right) dx = 2\sin\frac{x}{2} + \sin 2x + c$

(g) $\int\left(\sqrt{x} - \frac{1}{2}\cos x\right) dx = \frac{2}{3}\sqrt{x^3} - \frac{1}{2}\sin x + c$

(h) $\int\left(x^2 + \sin 2x\right) dx = \frac{x^3}{3} - \frac{1}{2}\cos 2x + c$

Week 5D

Key Idea – Simple Substitution

1. (a)

$$\int 2x(x^2-1)^4\ dx$$

Step 1: Let $u = x^2 - 1$

Step 2: $\dfrac{du}{dx} = 2x$ and $du = 2xdx$

Step 3: $\displaystyle\int 2x(x^2-1)^4\ dx = \int (x^2-1)^4\ 2xdx$

$\displaystyle = \int u^4 du$

Step 4: $\displaystyle = \frac{u^5}{5} + c$

Step 5: $\displaystyle = \frac{\left(x^2-1\right)^5}{5} + C$

(b)

$$\int 3x^2(x^3+5)^4\ dx$$

Step 1: Let $u = x^3 + 5$

Step 2: $\dfrac{du}{dx} = 3x^2$ and $du = 3x^2dx$

Step 3: $\displaystyle\int 3x^2(x^3+5)^4\ dx = \int (x^3+5)^4\ 3x^2dx$

$\displaystyle = \int u^4 du$

Step 4: $\displaystyle = \frac{u^5}{5} + c$

Step 5: $\displaystyle = \frac{\left(x^3+5\right)^5}{5} + c$

(c)

$$\int x^6\left(x^7-3\right)^5\ dx$$

Step 1: Let $u = x^7 - 3$

Step 2: $\dfrac{du}{dx} = 7x^6$ and $\dfrac{du}{7} = x^6dx$

Step 3: $\displaystyle\int x^6\left(x^7-3\right)^5\ dx = \int u^5\ \frac{du}{7}$

$\displaystyle = \frac{1}{7}\int u^5\ du$

Step 4: $\displaystyle = \frac{1}{7} \times \frac{u^6}{6} + c$

$\displaystyle = \frac{u^6}{42} + c$

Step 5: $\displaystyle = \frac{\left(x^7-3\right)^6}{42} + c$

(d)

$$\int \left(3-2x\right)^6\ dx$$

Step 1: Let $u = 3 - 2x$

Step 2: $\dfrac{du}{dx} = -2x$ and $\dfrac{du}{-2} = xdx$

Step 3: $\displaystyle\int \left(3-2x\right)^6\ dx = \int u^6\ \frac{du}{-2}$

$\displaystyle = -\frac{1}{2}\int u^6\ du$

Step 4: $\displaystyle = -\frac{1}{2} \times \frac{u^7}{7} + c$

$\displaystyle = -\frac{u^7}{14} + c$

Step 5: $\displaystyle = -\frac{\left(3-2x\right)^7}{14} + c$

(e)

$$\int \frac{2x+4}{\left(x^2+4x+7\right)^5}\ dx$$

Step 1: Let $u = x^2+4x+7$

Step 2: $\dfrac{du}{dx} = 2x+4$ and $du = \left(2x+4\right)dx$

Step 3: $\displaystyle\int \frac{2x+4}{\left(x^2+4x+7\right)^5}\ dx \ = \int \frac{1}{\left(x^2+4x+7\right)^5}(2x+4)\ dx$

$$= \int \frac{1}{u^5}\ du$$

$$= \int u^{-5} du$$

Step 4: $\displaystyle = \frac{u^{-4}}{-4} + C$

Step 5: $\displaystyle = \frac{\left(x^2+4x+7\right)^{-4}}{-4} + c$

$$= -\frac{1}{4\left(x^2+4x+7\right)^4} + C$$

(f)

$$\int \frac{1}{\left(2x+3\right)^3}\ dx$$

Step 1: Let $u = 2x+3$

Step 2: $\dfrac{du}{dx} = 2$ and $\dfrac{du}{2} = dx$

Step 3: $\displaystyle\int \frac{1}{\left(2x+3\right)^3}\ dx \ = \int \frac{1}{u^3}\ \frac{du}{2}$

$$= \frac{1}{2}\int u^{-3} du$$

Step 4: $\displaystyle = \frac{1}{2}\frac{u^{-2}}{-2} + c$

Step 5: $\displaystyle = \frac{\left(2x+3\right)^{-2}}{-4} + c$

$$= -\frac{1}{4\left(2x+3\right)^2} + c$$

(g)

$$\int 2x\sqrt[3]{x^2+1}\ dx$$

Step 1: Let $u = x^2+1$

Step 2: $\dfrac{du}{dx} = 2x$ and $du = 2xdx$

Step 3: $\displaystyle\int 2x\sqrt[3]{x^2+1}\ dx \ = \int 2x\left(x^2+1\right)^{1/3}\ dx$

$$= \int u^{1/3}\ du$$

$$= \frac{u^{4/3}}{4/3} + c$$

Step 4: $\displaystyle = \frac{3u^{4/3}}{4} + c$

Step 5: $\displaystyle = \frac{3(x^2+1)^{4/3}}{4} + c \ \text{ or } \ \frac{3}{4}(x^2+1)^{4/3} + c$

$$= \frac{3\sqrt[3]{(x^2+1)^4}}{4} + c \ \text{ or } \ \frac{3}{4}\sqrt[3]{(x^2+1)^4} + c$$

(h)

$$\int 2x\sqrt{1-x^2}\ dx$$

Step 1: Let $u = 1-x^2$

Step 2: $\dfrac{du}{dx} = -2x$ and $\dfrac{du}{-1} = 2xdx$

Step 3: $\displaystyle\int 2x\sqrt{1-x^2}\ dx \ = \int \sqrt{u}\ \frac{du}{-1}$

$$= (-1)\int u^{1/2} du$$

Step 4: $\displaystyle = (-1)\frac{u^{3/2}}{3/2} + c$

Step 5: $\displaystyle = (-1)\frac{2\left(1-x^2\right)^{3/2}}{3} + c$

$$= \frac{-2\left(1-x^2\right)^{3/2}}{3} + c \ \text{ or } \ \frac{-2}{3}\left(1-x^2\right)^{3/2} + c$$

$$= \frac{-2\left(\sqrt[2]{(1-x^2)^3}\right)}{3} + c \ \text{ or } \ \frac{-2}{3}\sqrt[2]{(1-x^2)^3} + c$$

(i) $9(3x-5)^{1/3} + c$ or $9\sqrt[3]{3x-5} + c$

(j) $(x^2-2x+3)^{1/2} + c$ or $\sqrt{(x^2-2x+3)} + c$

2. (a)

$$\int_0^1 5x^4\left(x^5-1\right)^3 dx$$

Step 1: Let $u = x^5 - 1$

Limits: For $x=1,\ u=1^5-1=0$

For $x=0,\ u=0^5-1=-1$

Step 2: $\dfrac{du}{dx}=5x^4$ and $du=5x^4dx$

Step 3: $\displaystyle\int_0^1 5x^4\left(x^5-1\right)^3 dx = \int_0^1 \left(x^5-1\right)^3 5x^4dx$

$$= \int_{-1}^0 u^3du$$

Step 4: $\displaystyle = \left[\frac{u^4}{4}\right]_{-1}^0$

$$= \frac{0^4}{4} - \frac{(-1)^4}{4}$$

$$= -\frac{1}{4}$$

(b)

$$\int_0^1 5x^4\left(x^5-1\right)^3 dx \quad \int_0^2 x^2(x^3+2)^3\ dx$$

Step 1: Let $u = x^3 + 2$

Limits: For $x=0,\ u=0^3+2=2$

For $x=2,\ u=2^3+2=10$

Step 2: $\dfrac{du}{dx}=3x^2$ and $\dfrac{du}{3}=x^2dx$

Step 3: $\displaystyle\int_0^2 x^2(x^3+2)^3\ dx = \int_2^{10} u^3\ \frac{du}{3}$

Step 4: $\displaystyle = \frac{1}{3}\left[\frac{u^4}{4}\right]_2^{10}$

$$= \frac{1}{12}\left[u^4\right]_2^{10}$$

$$= \frac{1}{12}\left(10^4 - 2^4\right)$$

$$= \frac{9984}{12}$$

$$= 832$$

(c)

$$\int_{-1}^1 \sqrt{1+x}dx = \int_{-1}^1 \left(1+x\right)^{1/2} dx$$

Step 1: Let $u = 1 + x$

Limits: For $x=-1,\ u=1+(-1)=0$

For $x=1,\ u=1+1=2$

Step 2: $\dfrac{du}{dx}=1$ and $du=dx$

Step 3: $\displaystyle\int_{-1}^1 \sqrt{1+x}dx = \int_0^2 u^{1/2}du$

Step 4: $\displaystyle = \left[\frac{u^{3/2}}{3/2}\right]_0^2$

$$= \frac{2}{3}\left[u^{3/2}\right]_0^2$$

$$= \frac{2}{3}\left(2^{3/2} - 0^{3/2}\right)$$

$$= \frac{2\sqrt{8}}{3}$$

$$= \frac{4\sqrt{2}}{3}$$

(d)

$$\int_0^2 \frac{1}{(3x-5)^{\frac{2}{3}}}dx = \int_0^2 (3x-5)^{-2/3}\ dx$$

Step 1: Let $u = (3x-5)$

Limits: For $x=0,\ u=3\times 0-5=-5$

For $x=2,\ u=3\times 2-5=1$

Step 2: $\dfrac{du}{dx}=3$ and $\dfrac{du}{3}=dx$

Step 3: $\displaystyle\int_0^2 \frac{1}{(3x-5)^{\frac{2}{3}}}dx = \int_{-5}^1 u^{-2/3}\ \frac{du}{3}$

Step 4: $\displaystyle = \frac{1}{3}\left[\frac{u^{1/3}}{1/3}\right]_{-5}^1$

$$= \left[u^{1/3}\right]_{-5}^1$$

$$= \left(1^{1/3} - (-5)^{1/3}\right)$$

$$= 1 + \sqrt[3]{5}$$